

RETAIL SALES PERFORMANCE ANALYSIS

Business Intelligence & Machine Learning Project

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PROJECT OBJECTIVE

As a Business Information Systems student, I wanted to go beyond theory and apply real analytical tools to a business problem that matters — understanding what actually drives profitability. Using a retail sales dataset of **500 orders** spanning two years, I built an end-to-end analysis combining Power BI dashboarding with three machine learning models in RapidMiner Studio, translating raw transaction data into actionable business intelligence across sales, discount strategy, and customer segmentation.

KEY METRICS AT A GLANCE

\$492K Total Revenue	\$49.4K Total Profit	34.4% Loss-Making Orders	500 Orders Analyzed
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DATASET

Source: Superstore Retail Sales Dataset (2022–2023) · **Records:** 500 orders · **Fields:** Order Date, Customer ID, Segment, Region, Category, Sub-Category, Sales, Quantity, Discount, Profit

1 · POWER BI DASHBOARD

Data Preparation

- Removed duplicates, corrected data types, resolved missing values
- Computed columns: Profit Margin % and Order Month Number
- Defined relationships across date, segment, and category dimensions

Dashboard Visuals (6+)

Visual	Type
Sales by Category	Bar Chart
Sales Trend Over Months	Line Chart
Sales by Segment	Donut Chart
Discount vs. Profit	Scatter Chart
Profit by Category	Donut Chart
Total Profit vs. Target	Gauge

Interactivity

Dynamic slicers for Order Month and Product Category allow real-time cross-filtering across all visuals simultaneously.

2 · MACHINE LEARNING — RapidMiner

A · Clustering (K-Means)

Features: Sales · Profit · Quantity · Discount

Output: 3 distinct customer value tiers from 500 orders:

Cluster	Size	Profile
High	123	High sales & profit
Medium	131	Moderate performance
Low	246	Low sales & profit

B · Regression (Linear)

Target: Profit **Features:** Sales, Quantity, Discount

Metrics: RMSE · MAE · R² evaluated on held-out test split. Model confirms sales and quantity positively affect profit; higher discounts systematically reduce it.

C · Classification (Decision Tree)

Target: Profit_Class — derived label (High / Medium / Low)

Features: Sales, Quantity, Discount

Metrics: Accuracy · Precision · Recall · F1-Score. Enables proactive flagging of low-profit orders before fulfillment.

KEY FINDINGS

Hidden Margin Losses

172 of 500 orders (34.4%) were loss-making, generating -\$16,770 in absorbed losses. Without a classification model to flag these proactively, this erosion goes undetected until year-end reporting.

Discount Danger Zone

Orders with discounts ≥20% carried a 35.2% loss rate versus 31.9% for undiscounted orders. Aggressive discounting is a measurable profit drain, not just a pricing preference.

Revenue ≠ Profitability

Consumer segment led revenue at \$192.6K, yet the Central region delivered the highest profit at \$16,511. Chasing volume without segment-level profit tracking produces misleading signals.

Segmentation Opportunity

Nearly half the customer base (246 of 500) sits in the low-value cluster — signaling that retention and upselling strategies should take priority over broad acquisition spend.

TOOLS & TECHNOLOGIES

Power BI

RapidMiner Studio

K-Means Clustering

Decision Tree

Linear Regression

CSV / Excel

DAX

CONCLUSION

This project demonstrates how combining **interactive BI dashboards** with **predictive machine learning** unlocks a layer of insight that neither tool delivers alone. Power BI exposed the sales trends and category gaps; the ML models went further — clustering customers by value, predicting profit outcomes, and classifying orders before they become losses. Together, they provide a decision-support framework with immediate business applicability: **revise discount thresholds, prioritize high-value customer retention, and redirect marketing spend from volume to margin.**